Ajinkya Taranekar

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Experience

Apple — **Dublin, Ireland**
Site Reliability Engineering Intern, ASE Security SRE Team — *Jun 2026 – Sept 2026*

- Support site reliability for security-critical distributed systems, handling monitoring, alerting, root-cause incident response, and automation to keep systems highly available.

Nuclei, CDNA Technologies Pvt. Ltd. — **Bengaluru, India**
Software Development Engineer 2 — *Jul 2024 – Jul 2025*

- Slashed partner bank onboarding time from weeks to under 5 minutes by designing a Go + gRPC Agent Management System with OIDC multi-tenant auth and Protobuf contracts for an AI-driven VoiceBot SDK, rolled out across multiple bank integrations.
- Reduced product catalogue search latency from 1s to 150ms by combining Milvus vector search with Elasticsearch fuzzy matching in a custom hybrid search pipeline.
- Powered proactive, data-driven decisions for 100K+ daily active users with a real-time analytics engine on Kafka and ClickHouse, processing metrics and GDPR RoPA events.
- Shipped an internal Figma plugin that uses LLMs to generate UX heatmaps and optimise copy, cutting design review cycles from weeks to days.
- Mentored a team of engineers in Java backend and Svelte frontend development.

Nuclei, CDNA Technologies Pvt. Ltd. — **Bengaluru, India**
Full-Stack Developer — *Jul 2022 – Jul 2024*

- Resolved 1,400+ P0 production outages independently at a 91.7% resolution rate (avg. ack 2.8 min) as the company's founding SRE, automating CIS Benchmark hardening, orchestrating Airflow DAGs for stuck-event recovery, and driving RCA on partner-facing payments and business services.
- Compressed new partner onboarding time from weeks to 1 week for India's largest banks (India & MEA) by automating the rollout of 300+ products across 15 categories into a multi-tenant CMS built on Spring Boot and gRPC.
- Migrated marketplace analytics from MySQL to ClickHouse with Airflow-orchestrated ETL, cutting report generation from hours to 30 seconds and letting teams build their own self-serve Superset dashboards.
- Trimmed frontend dev time by 75% across 6 banking categories by building a core NPM package and leading 6 engineers through the Merchant Marketplace's migration to a Svelte micro-frontend architecture with atomic design patterns.
- Built the Checkout System handling 100,000+ daily transactions, with multi-currency formatting, discount coupons, cart expiry logic, and Wallet/Coupon payments on Svelte and Spring Boot.

Projects

Constitutional AI Framework on Small Language Model — **GitHub**
FastAPI, Python, PyTorch, Unsloth, Qwen3, HuggingFace — *Sept 2025 – Aug 2026*

- Fine-tuned a Qwen3-0.6B PyTorch LLM using Unsloth's 16-bit LoRA on synthetic constitutional data to test how alignment reliability holds up at on-device scale.
- Developed a modular Python reliability harness, combining a constitutional scratchpad, a 5W+H user memory model, and a dependency monitor, to enable multi-turn personalization without retraining the model.
- Designed a dual-SFT Thinker-Executor architecture that splits reasoning and tool-calling across two sub-1B models, keeping everything on-device to preserve data privacy.

Atlas: ML Training Reliability & Observability | HackEurope 2026, Best Use of Crusoe API — **Devpost**
FastAPI, Python, PyTorch, Next.js, SQLite, Crusoe Cloud, Anthropic Claude, WebGL, CodeCarbon, MCP — *Feb 2026*

- Engineered a real-time observability platform for PyTorch training runs; a custom Python Observer library streams per-epoch telemetry to a FastAPI backend that runs 30+ diagnostic heuristics, covering layer health, gradient flow, architectural waste, early stopping, device utilisation, and carbon emissions.
- Extended the diagnostics engine with an agentic AI pipeline of LLM-powered agents, tracked CO_2 emissions via CodeCarbon using EU ETS pricing, and exposed the tools through an MCP server for Claude Desktop.

PG AI: PostgreSQL AI Toolkit Extension — **GitHub**
PostgreSQL, C++, OpenAI API, Anthropic API, OpenRouter — *Nov 2025*

- Created a PostgreSQL extension for CLI-level AI tooling: natural-language-to-SQL queries via `ai_toolkit.query(...)`, query explanation via `explain_query(...)`, and error diagnosis via `explain_error(...)`.
- Supported multi-provider LLM config (OpenAI, Anthropic, OpenRouter) with persistent, schema-aware "database memory" for context-aware query generation.

Technical Skills

Languages: Python, Java, Go, TypeScript, JavaScript, C++, Dart, SQL

Backend & Distributed Systems: FastAPI, Spring Boot, gRPC/Protobuf, REST APIs, Microservices, Kafka, Redis, Airflow, Apache NiFi

DevOps & Cloud: Docker, Docker Swarm, Kubernetes, Helm, Terraform, Azure, GCP, GitHub Actions, Istio, Jenkins, CI/CD, DevSecOps

Security & Identity: OIDC/OAuth2.0, IAM Architecture, HashiCorp Vault, Secret Management, Supply Chain Security, CIS Benchmark Hardening, Multi-Tenant Access Control

Observability & Reliability: Grafana, Prometheus, OpenSearch, ELK Stack, Superset, GoAlert, PMM, Slack Alerts, SLA/SLO

Data & Storage: PostgreSQL, MySQL, CockroachDB, ClickHouse, Elasticsearch, Milvus, FalkorDB, SQLite

AI & LLMs: LangChain, RAG, Vector Search, Chunking, Prompt/Context Engineering, MCP, PyTorch Fine-Tuning (LoRA/Unsloth), Multi-Agent Orchestration, Harness Engineering, LLM APIs (Claude, GPT, Qwen)

Frontend: React, Svelte, Next.js, Flutter, Micro-Frontends, NPM Packaging, Atomic Design Systems, Figma Plugins

Education

Trinity College Dublin: MSc Computer Science, Future Networked Systems, 1:1, Distinction — *Dublin, Ireland | Sept 2025 – Sept 2026*

Shri G.S. Institute of Technology and Science: B.Tech CS, CGPA 9.04/10.0, Rank 1st, Gold Medal — *Indore, India | Aug 2018 – Jun 2022*